
Handout - *Rudolph Carnap, “Empiricism, Semantics, and Ontology”*

The central problem Rudolf Carnap addresses is the long-standing philosophical dispute over *abstract entities*—numbers, properties, propositions, classes—and whether reference to such entities commits one to a *metaphysical* (often labeled “Platonic”) ontology incompatible with empiricism. Empiricists and nominalists worry that admitting abstract entities violates scientific rigor and reintroduces metaphysics through the back door, especially in mathematics, physics, and (most pressingly for Carnap) **semantics**.

Carnap’s proposed solution is to dissolve the dispute by drawing a sharp distinction between *internal* questions (asked *within* a linguistic framework) and *external* questions (asked *about* the framework as a whole). Once this distinction is in place, Carnap argues, traditional ontological debates about the “reality” of abstract entities are exposed as *pseudo-questions*. Accepting abstract entities is not a metaphysical thesis but a *practical decision* to adopt certain linguistic forms, justified by considerations of *expediency, fruitfulness, and simplicity*.

I. Historical and Philosophical Context (Supplement)

Carnap’s essay (originally published in the 1950s) emerges from **logical empiricism**, especially the Vienna Circle (Schlick, Carnap, Neurath), which sought to eliminate metaphysics by subjecting philosophical claims to logical and linguistic analysis. Influences include:

- **Wittgenstein (Tractatus)**: The idea that many philosophical problems arise from misuse of language.
- **Frege and Russell**: The development of modern logic and logicism, which legitimized quantification over abstract objects.
- **Nominalism vs. Realism**: Medieval debates about universals resurface in 20th-century analytic philosophy, now reframed in terms of logic and semantics.
- **Quine**: Quine’s criterion (i.e. *to be is to be the value of a bound variable*) looms large (see footnote 3).

Carnap’s intervention aims not to *win* the ontological debate but to *reframe* it as a matter of linguistic engineering rather than metaphysical discovery.

II. Linguistic Frameworks and the Internal/External Distinction

Carnap introduces the notion of a **linguistic framework**: a system of expressions governed by rules that allow us to speak meaningfully about a certain kind of entity.

A. Internal Questions

- Asked *within* a framework.
- Answerable by **empirical methods** (for factual frameworks) or **logical/analytic methods** (for logical frameworks).
- Examples:
 - *Thing-language*: “Is there a white piece of paper on my desk?” (empirical)
 - *Number framework*: “Is there a prime number greater than 100?” (analytic)

Internal existence claims (e.g., “There are numbers”) are often **analytic and trivial**, once the framework is in place (pp. 14–15).

B. External Questions

- Asked *about* the framework as a whole.
- Examples: “Are there really numbers?” “Is the external world real?”
- Carnap argues these are **non-cognitive pseudo-questions** when construed as theoretical assertions (pp. 14–16).

Instead, external questions are best understood as *practical decisions* about whether to adopt a framework, guided by pragmatic criteria rather than truth-values.

III. Case Studies of Frameworks (Structure Mirroring the Text)

Carnap systematically applies the internal/external distinction across domains.

A. The Framework of Things (pp. 14–15)

- Acceptance of *thing-language* allows empirical inquiry.
- Asking whether the *thing world itself* is real is a misuse of the concept of reality.

- To accept the thing world = to accept rules for forming and testing statements, **not** to affirm a metaphysical belief.

B. The Framework of Numbers (pp. 15–16)

- Numbers are introduced via numerals, predicates (“is a number”), variables, and quantifiers.
- Internal questions are analytic.
- External ontological debates about numbers’ “independent existence” lack cognitive content.

C. Propositions, Properties, and Universals (pp. 16–18)

- Propositions are introduced as values of sentential variables.
- Carnap argues they are **not** mental, linguistic, or subjective entities—this follows from their logical role.
- External realism/nominalism debates about universals are again pseudo-questions.

D. Real Numbers and Space-Time Points (pp. 17–18)

- Adoption of real numbers and space-time coordinates in physics is a **pragmatic choice**, motivated by mathematical simplicity and explanatory power.
- External questions like “Are space-time points real?” are either trivial (internal) or misguided (external).

IV. What Acceptance of a Kind of Entities Means

Carnap clarifies that accepting a framework involves:

1. Introducing a **general predicate** for the entity type (e.g., “number,” “property”).
2. Introducing **variables** ranging over those entities.

This does *not* require introducing names for entities, nor does it assert some sort of metaphysical reality (pp. 17–18).

Key claim: Acceptance is neither true nor false—it is a *decision*, evaluable only in terms of utility.

Carnap criticizes the tendency (especially among nominalists) to label this acceptance as “Platonism,” calling it deeply misleading (p. 18).

V. Abstract Entities in Semantics

For Carnap, the flashpoint is **semantics**, where expressions are said to *designate* entities.

A. The Nominalist Objection

Critics reject statements like:

- “The word ‘five’ designates a number.”
- “The sentence ‘Chicago is large’ designates a proposition.”

They accuse semanticists of *hypostatization* (Ryle’s “Fido–Fido fallacy”).

B. Carnap’s Reply (pp. 18–20)

- If one accepts a framework, then designation claims are **analytic truths** within it.
- The dispute again confuses internal assertions with external acceptance.
- Demanding *evidence* for abstract entities mistakes acceptance of a framework for a factual hypothesis.

Carnap argues that no shared notion of evidence could resolve such disputes, confirming their pseudo-question status.

VI. Conclusion: Methodological Tolerance

Carnap closes with a plea for **tolerance and experimentalism** in linguistic and scientific practice (pp. 20–21):

- The real question is not “Do abstract entities exist?” but “Are these linguistic tools useful?”
- History shows that dogmatic prohibitions on forms of expression hinder scientific progress.